

What is a function?

collections of things

- Take two sets D and E

a function is a rule/map that assigns at most 1 object in E for each object in D

called $f(x)$

called x

$f: D \rightarrow E$, D is called the domain
 E is called the co-domain

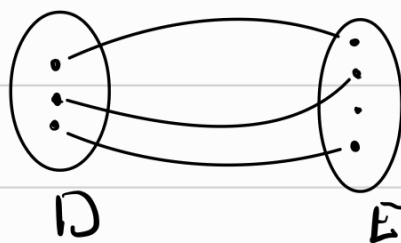
Domain: all possible inputs

Range: all possible outputs from Domain.

Some classes of functions

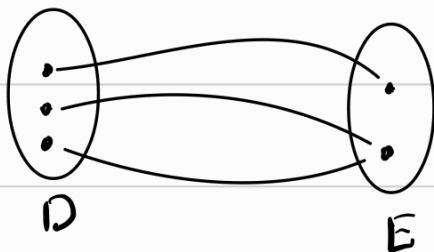
• One-to-One (injective)

Each value in D goes to ^{different} some value in E



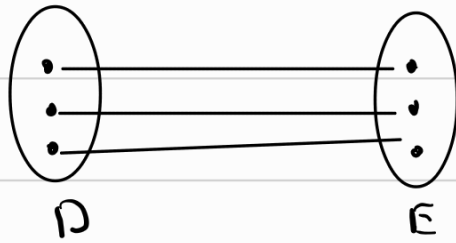
• Onto (surjective)

Each value in E is mapped to



• Bijjective

Both onto and one-to-one

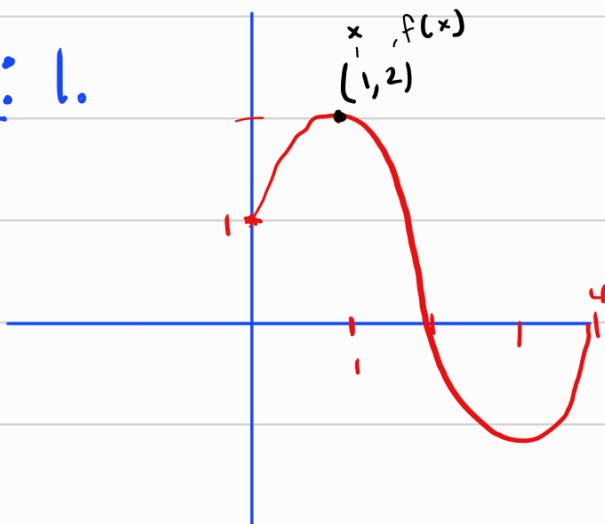


In a sense, just think of functions as machines that take inputs to outputs.

4 ways to represent functions

1. Verbally — describing it in words
2. Numerically — table of values
- ★ 3. Visually — graph
- ★ 4. Algebraically — a formula

Ex: 1.



Domain is all inputs: $[0, 4]$

Range is all outputs: $[-1, 2]$

2.

$$f(x) = x^2$$

Domain: $\mathbb{R}$

Range: $[0, \infty)$

3. $f(x) = 2x - 1$

D: $\mathbb{R}$

R: $\mathbb{R}$

4. (Practice with composition)

$$f(x) = x^2, \quad \text{find} \quad \frac{f(x+h) - f(x)}{h}$$

5. $f(x) = \sqrt{x+2}$

$$D = [-2, \infty)$$

6. $f(x) = \frac{1}{x^2 - x} = \frac{1}{x(x-1)}$

$$D = (-\infty, 0) \cup (0, 1) \cup (1, \infty)$$

How to tell when something is a function

Functions abide by one rule:

"For every input, there is at most one output"

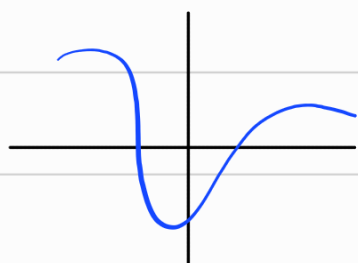
For example, the following does not represent a function of x to y

x	2	4	5	5	6
y	3	6	7	8	9

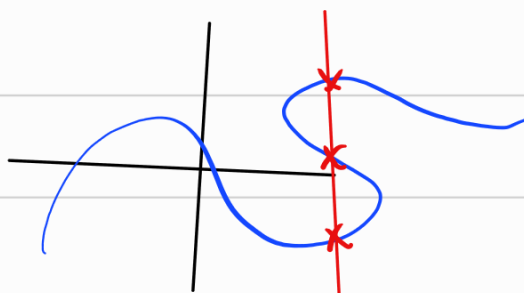
as an input of 5 has two different outputs

This is easy to check on a graph via the
Vertical line test

↳ draw vertical line, if there are more than one intersection, then the graph doesn't represent a function.



Function



Not a function

Piecewise Defined Functions

Sometimes it is useful to define a function in multiple parts. When this is done, it's called a Piecewise function

They look as follows

$$f(x) = \begin{cases} \sim & \text{Interval 1} \\ \sim & \text{Interval 2} \\ \vdots & \\ \sim & \text{Interval } n \end{cases}$$

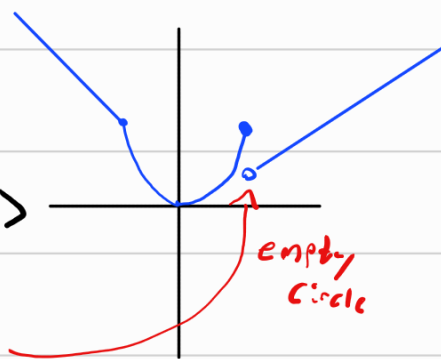
↳ expressions or formulas

For example

$$f(x) = \begin{cases} -x & x < 0 \\ x & x \geq 0 \end{cases}$$



$$f(x) = \begin{cases} -x+2 & x \leq -1 \\ x^2 & -1 < x \leq 1 \\ \frac{1}{2}x & 1 < x \end{cases}$$



Note: When creating these functions, be careful to ensure the intervals do not overlap as these are still functions

Even & Odd functions

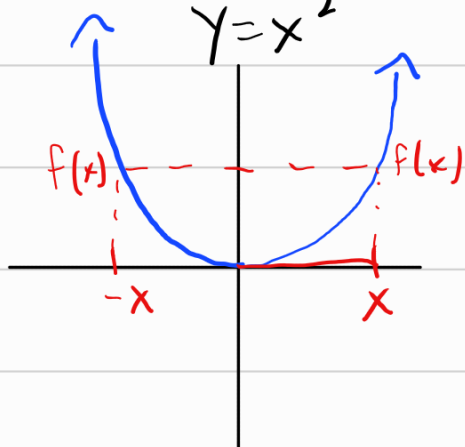
Def: A function is even if and only if $f(x) = f(-x)$ for all x in the domain.

Def: A function is odd if and only if $-f(x) = f(-x)$ for all x in the domain.

Example

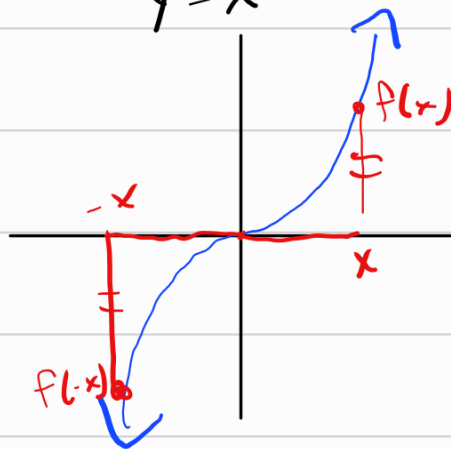
Even

$$y = x^2$$



odd

$$y = x^3$$

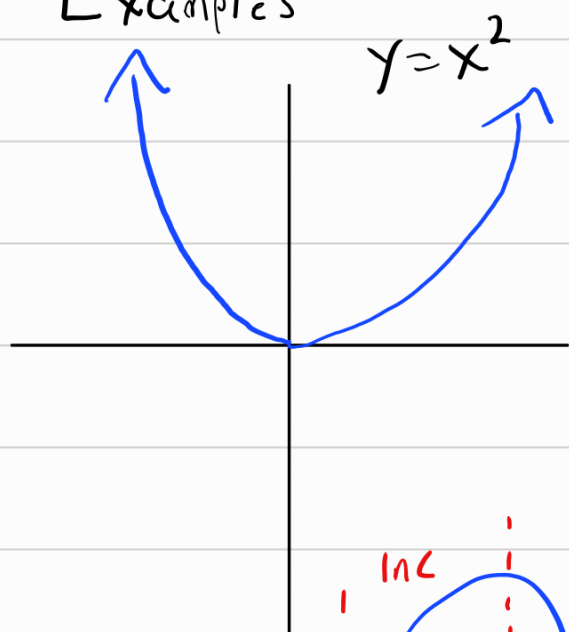


Increasing & Decreasing

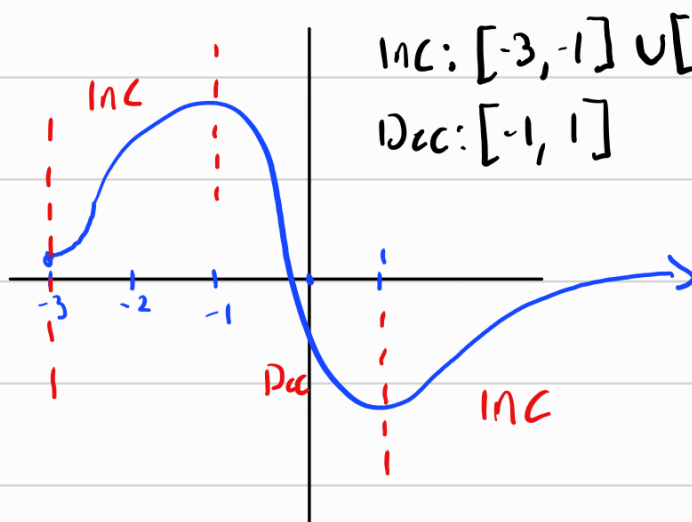
Def: A function is increasing on an interval $[a, b]$ if for $x_1, x_2 \in [a, b]$ where $x_1 < x_2$ we have $f(x_1) \leq f(x_2)$

Def: A function is decreasing on an interval $[a, b]$ if for $x_1, x_2 \in [a, b]$ where $x_1 < x_2$ we have $f(x_1) \geq f(x_2)$

Examples



$\leadsto$ Increasing on $[0, \infty)$
Decreasing on $(-\infty, 0]$



INC: $[-3, -1] \cup [1, \infty)$
DEC: $[-1, 1]$